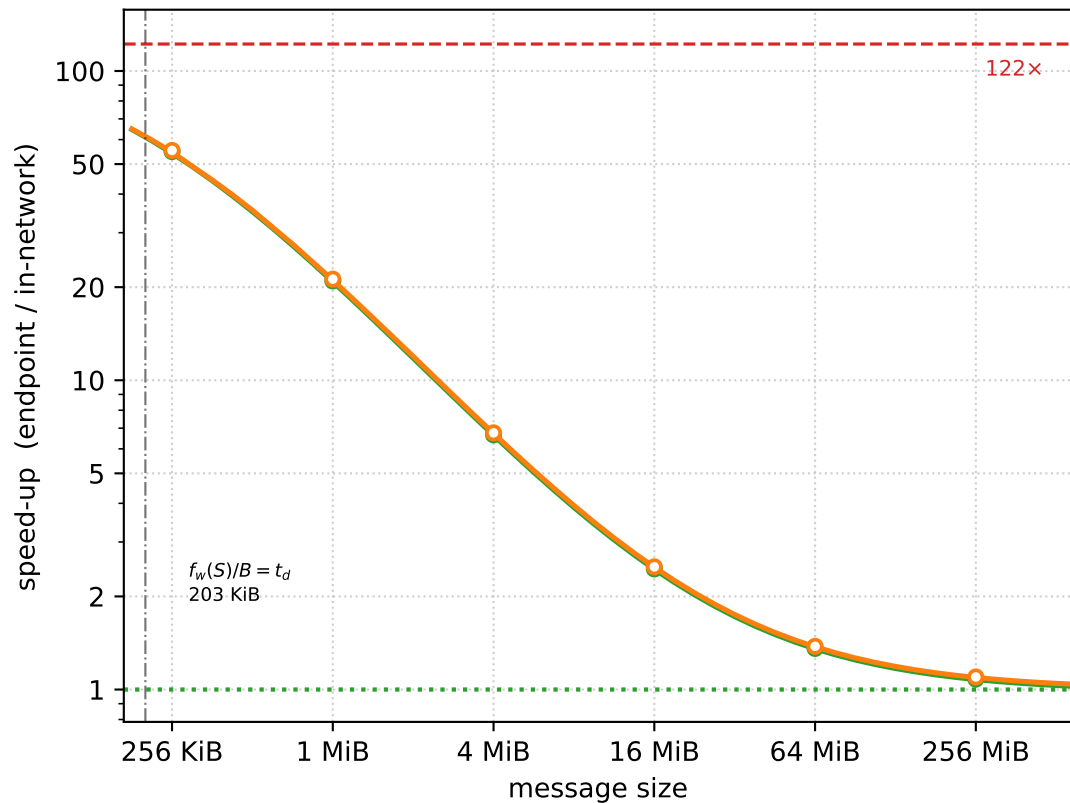
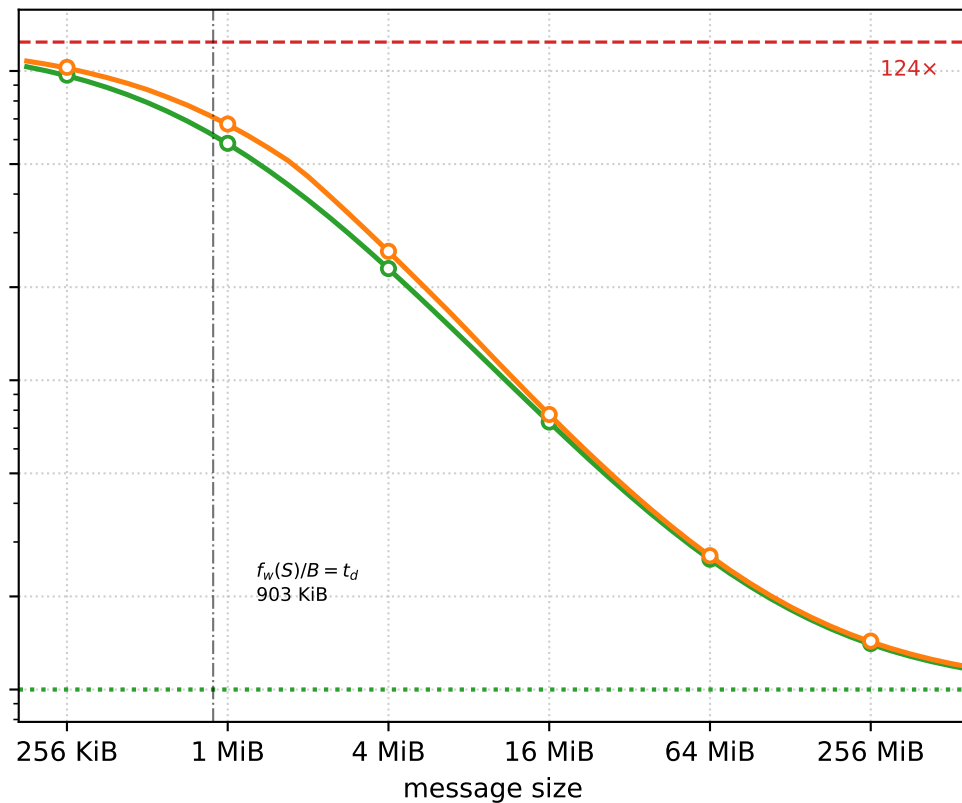


(a) single switch, $d = 1$ (b) three-tier, $d = 3$ 

— Reduce-Scatter, model

— AllGather, model

- - - latency bound $(N-1)\lambda_{\max}/t_d$

○ Reduce-Scatter, measured

○ AllGather, measured

... bandwidth bound 1 (AG); $(N-1)/N$ (RS)